

CONTEST QUESTION PAPER

Contest ID: 21

Total Questions: 37 | Each question carries equal marks

Instructions: Choose the best answer from the options given below each question.

Name: _____ Roll No: _____ Date: _____

Section A: Web Technologies (HTML, CSS, JavaScript)

Q1. Which HTML tag is used to define an internal style sheet?

- (A) <style>
- (B) <css>
- (C) <script>
- (D) <link>

Answer: _____

Q2. What does the 'alt' attribute in an tag do?

- (A) Adds a border to the image
- (B) Provides alternative text if the image cannot be displayed
- (C) Sets the image alignment
- (D) Changes image size

Answer: _____

Q3. Which HTML5 tag is used to represent a self-contained piece of content?

- (A) <section>
- (B) <article>
- (C) <aside>
- (D) <main>

Answer: _____

Q4. What is the correct CSS syntax to change the text color of all <p> elements to red?

- (A) p {color: red;}
- (B) p {text-color: red;}
- (C) p {font-color: red;}
- (D) all {p: red;}

Answer: _____

Q5. Which CSS property is used to control the space between elements' borders and their content?

- (A) margin
- (B) padding
- (C) spacing
- (D) border

Answer: _____

Q6. What does 'em' unit represent in CSS?

- (A) Pixels relative to screen
- (B) Font size of the root element
- (C) Font size of the current element

(D) Percentage of the viewport

Answer: _____

Q7. Which CSS property is used to make a flex container?

(A) display: flex

(B) float: flex

(C) position: flex

(D) layout: flex

Answer: _____

Q8. What is the output of `typeof null` in JavaScript?

(A) null

(B) object

(C) undefined

(D) string

Answer: _____

Q9. Which method removes the last element from a JavaScript array and returns it?

(A) shift()

(B) pop()

(C) splice()

(D) slice()

Answer: _____

Q10. What does `===` check in JavaScript?

(A) Only value equality

(B) Only type equality

(C) Both value and type equality

(D) Neither value nor type

Answer: _____

Q11. What is a closure in JavaScript?

(A) A function that is immediately invoked

(B) A function that has access to its outer function's scope even after the outer function returns

(C) A method to close a browser window

(D) A way to stop a loop

Answer: _____

Section B: React.js

Q12. What is the purpose of the `useEffect` hook in React?

(A) To manage component state

(B) To perform side effects in functional components

(C) To create new components

(D) To handle routing

Answer: _____

Q13. What is JSX in React?

(A) A JavaScript XML library

(B) A syntax extension that allows writing HTML-like code in JavaScript

(C) A new version of JavaScript

(D) A CSS framework for React

Answer: _____

Q14. Which of the following correctly describes props in React?

- (A) Mutable data managed inside a component
- (B) Immutable data passed from parent to child component
- (C) A lifecycle method
- (D) A type of React hook

Answer: _____

Q15. What is the virtual DOM in React?

- (A) A direct copy of the real DOM
- (B) A lightweight JavaScript representation of the real DOM used for efficient updates
- (C) A browser API
- (D) A server-side rendering technique

Answer: _____

Q16. What does the useState hook return in React?

- (A) Only the state value
- (B) Only the setter function
- (C) An array with the state value and a function to update it
- (D) An object with state properties

Answer: _____

Q17. What is the purpose of the key prop when rendering lists in React?

- (A) It applies styles to list items
- (B) It helps React identify which items have changed, added or removed
- (C) It sets the order of list items
- (D) It is required for all elements

Answer: _____

Section C: Node.js, Express & MongoDB

Q18. Which Node.js module is used to create an HTTP server?

- (A) url
- (B) http
- (C) path
- (D) fs

Answer: _____

Q19. What does npm stand for?

- (A) Node Package Manager
- (B) New Project Module
- (C) Node Program Manager
- (D) None of the above

Answer: _____

Q20. What is middleware in Express.js?

- (A) A database connection layer
- (B) A function that has access to the request and response objects and can modify them
- (C) A type of HTTP method

(D) A routing algorithm

Answer: _____

Q21. How do you define a route for HTTP GET requests in Express.js?

- (A) `app.post('/route')`
- (B) `app.get('/route', callback)`
- (C) `app.route.get('/path')`
- (D) `express.get('/path')`

Answer: _____

Q22. What is MongoDB?

- (A) A relational database management system
- (B) A NoSQL document-oriented database
- (C) A SQL-based database
- (D) A caching database only

Answer: _____

Q23. What is a replica set in MongoDB?

- (A) A backup copy of a collection
- (B) A group of mongod processes that maintain the same data set for redundancy and high availability
- (C) A type of index
- (D) A sharding strategy

Answer: _____

Q24. Which JavaScript method is used to handle a resolved Promise?

- (A) `then()`
- (B) `catch()`
- (C) `finally()`
- (D) `resolve()`

Answer: _____

Q25. What is event delegation in JavaScript?

- (A) Assigning multiple events to one element
- (B) Adding events to each child element individually
- (C) Attaching a single event listener to a parent element to handle events on child elements
- (D) Delegating events to the browser

Answer: _____

Q26. What does the spread operator (...) do in JavaScript?

- (A) Closes an array
- (B) Creates a deep copy always
- (C) Expands an iterable into individual elements
- (D) Merges two functions

Answer: _____

Q27. What is the difference between null and undefined in JavaScript?

- (A) They are identical
- (B) null is an assigned value representing no value; undefined means a variable has been declared but not assigned
- (C) undefined is always a bug
- (D) null cannot be used in conditions

Answer: _____

Q28. What is a MongoDB aggregation pipeline?

- (A) A method to insert bulk data
- (B) A framework to process data records through multiple stages like match, group, sort and project
- (C) A replication feature
- (D) An indexing strategy

Answer: _____

Section D: Aptitude & Reasoning

Q29. A man buys a watch for Rs.1200 and sells it for Rs.1500. What is the profit percentage?

- (A) 20%
- (B) 25%
- (C) 30%
- (D) 15%

Answer: _____

Q30. A train travels 360 km in 4 hours. What is its speed in km/h?

- (A) 80
- (B) 90
- (C) 100
- (D) 72

Answer: _____

Q31. If 8 workers can complete a task in 12 days, how many days will 16 workers take to complete the same task?

- (A) 4
- (B) 6
- (C) 8
- (D) 10

Answer: _____

Q32. A tank is filled by pipe A in 20 minutes and emptied by pipe B in 30 minutes. If both are open, how long to fill the tank?

- (A) 40 minutes
- (B) 50 minutes
- (C) 60 minutes
- (D) 45 minutes

Answer: _____

Q33. Arrange the following words in logical order: 1-Word, 2-Sentence, 3-Letter, 4-Paragraph

- (A) 3-1-2-4
- (B) 1-2-3-4
- (C) 3-2-1-4
- (D) 4-3-2-1

Answer: _____

Q34. If in a code MANGO is written as OCPIQ, what is the code for APPLE?

- (A) CRRNG
- (B) CRRNF
- (C) BQQMF

(D) DRRNH

Answer: _____

Q35. Pointing to a photograph Ram said 'She is the daughter of my grandfather's only son.' How is the girl related to Ram?

- (A) Sister
- (B) Daughter
- (C) Niece
- (D) Cousin

Answer: _____

Section E: Practical Coding Questions (JavaScript)

Q36. Write a JavaScript program to check whether a given number is an Armstrong number or not.

Input :- Number is 154 :

Q37. Write a JavaScript program to check whether a given year is a Leap Year or not.

Input : Year is 2019
